12.4 Independent in Pairs, Not All Together

WORKSHEET

The question

STEP III 2021, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What is $\mathrm{P}(X_{12} = 1)$? What is $\mathrm{P}(X_{12} = 1 \text{ and } X_{23} = 1)$?

STAGE 2 Mean and variance of the score

Ask yourself When does the variance of a sum equal the sum of the variances?

STAGE 4 The odd-and-even game

Ask yourself What is $\mathrm{E}(Z_{12})$? What values can $Z_{12}Z_{23}$ take, and when is it non-zero?

Try this next

1. **Two values.** (From STEP III, 2000.) X takes the values $x_1 \neq x_2$ and Y the values $y_1 \neq y_2$, with $\mathrm{P}(X = x_1, Y = y_1) = a$, $\mathrm{P}(X = x_1) = q$, $\mathrm{P}(Y = y_1) = p$. Show that if $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$ then $(a - pq)(x_1 - x_2)(y_1 - y_2) = 0$, and deduce that X and Y are independent. Give a joint distribution for A and B, each uniform on $\{-1, 0, 1\}$, with $\mathrm{E}(AB) = \mathrm{E}(A)\mathrm{E}(B)$ but not independent.

2. **A bigger team.** For $n = 7$ in part (i), find the mean and standard deviation of the score, and compare the example score 7 with them.

3. **Birthdays again.** Use indicators to find the mean and variance of the number of pairs sharing a birthday among k people (Section 9.9). Are the indicators pairwise independent?

4. **Pairwise but not triple.** Find three events, each of probability $\frac{1}{2}$, with $\mathrm{P}(A \cap B \cap C) = \frac{1}{8}$ but which are not pairwise independent.

Further reading

- **Geoffrey Grimmett & David Stirzaker, *Probability and Random Processes*** (Oxford University Press), the opening chapters on independence.
- **Michael Mitzenmacher & Eli Upfal, *Probability and Computing*** (Cambridge University Press). Pairwise independence and its uses in algorithms.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online), on indicators and covariance.

The full guided solution, and what the question leads to at university, is in Section 12.4:
https://beyondstep.keithmcnulty.org/ch12/pairwise-independence.html